

AI/ML Task-1: Student Performance Data Explorer

Exploratory Data Analysis Report

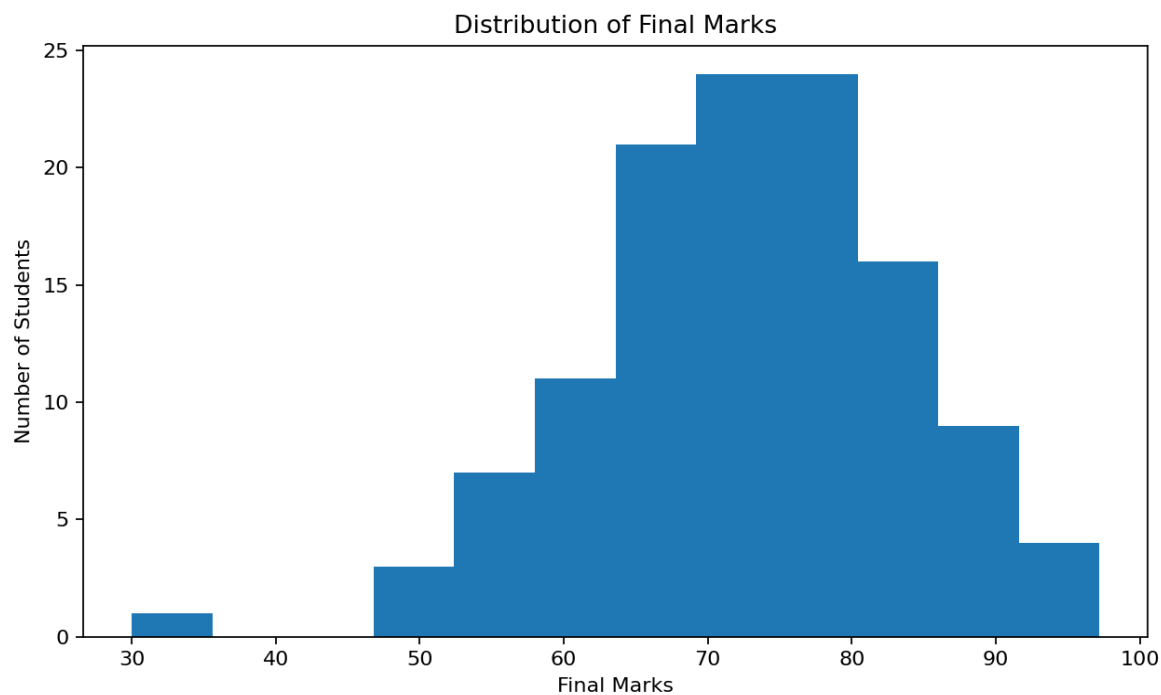
Objective: Import and clean student performance data, perform exploratory data analysis, generate charts, and summarize the findings.

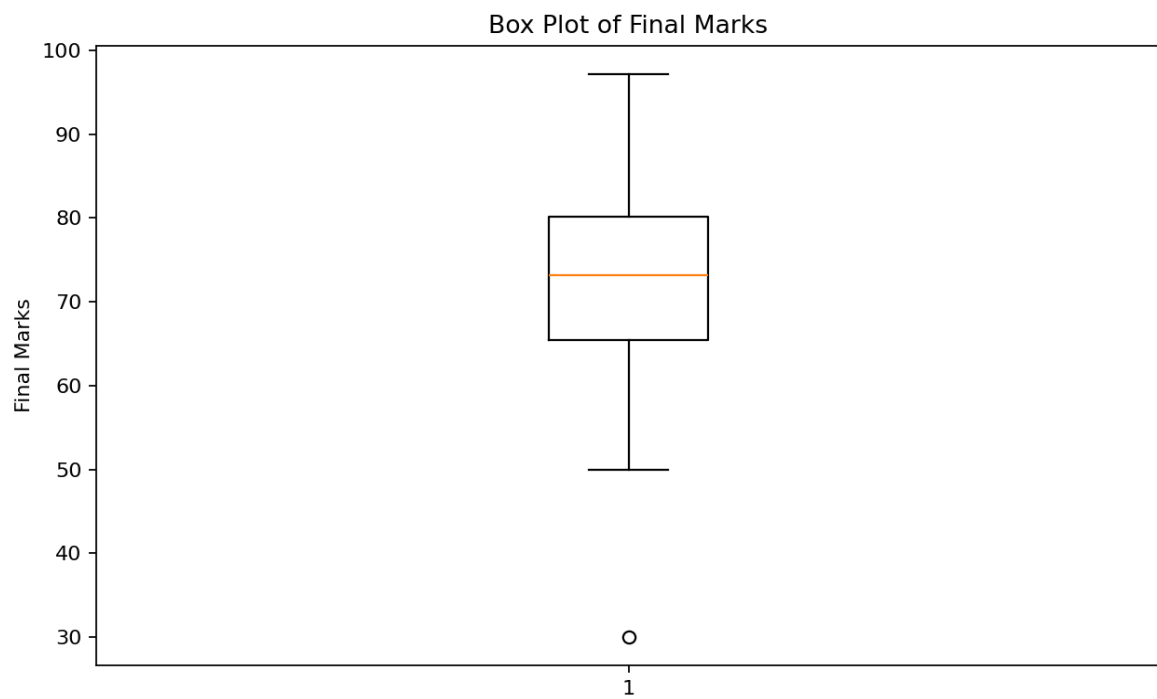
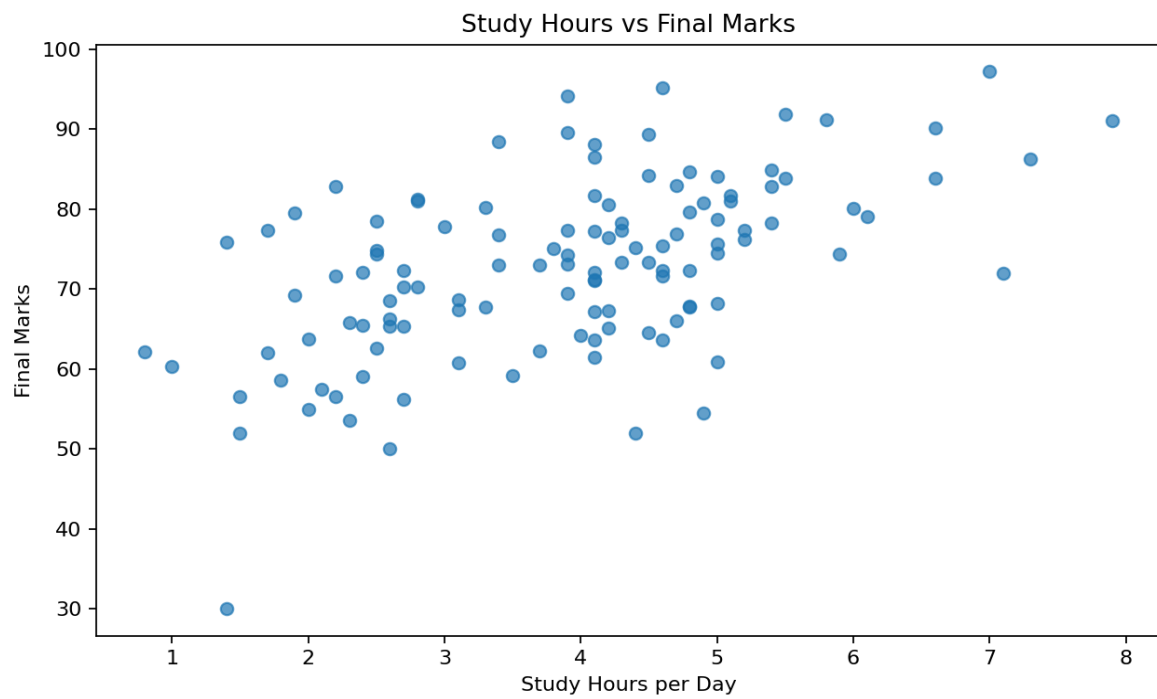
Dataset: 122 rows initially, including 2 duplicate records. After duplicate removal, 120 unique records remain. The dataset contains attendance, study hours, assignment score, internal marks and final marks.

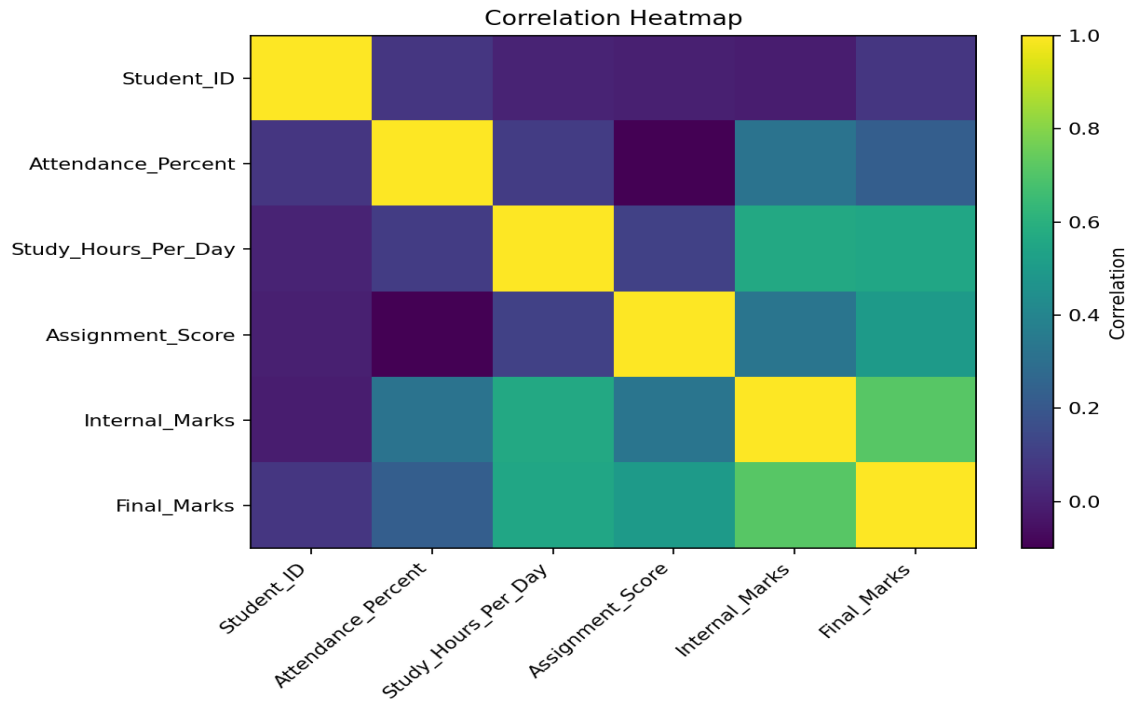
Cleaning: Duplicate records were removed. Missing numeric values in Study_Hours_Per_Day and Assignment_Score were replaced with their respective medians.

Metric	Value
Mean Final Marks	72.74
Median Final Marks	73.20
Minimum Final Marks	30.0
Maximum Final Marks	97.2

Visualizations







Key Findings

1. The dataset is suitable for practicing the complete EDA workflow.
2. Cleaning improves the dataset by removing duplicate observations and filling missing numeric values.
3. The histogram shows the distribution of final marks.
4. The scatter plot provides a visual view of the relationship between daily study hours and final marks.
5. The correlation matrix helps identify relationships among numeric variables.
6. The cleaned dataset can be carried forward into Task 2 for machine-learning preprocessing and baseline modeling.

Conclusion

Task 1 successfully demonstrates data loading, inspection, cleaning, descriptive statistics, visualization and documentation. The project follows the requested Student Performance Data Explorer workflow.